Tunneling+ v1

A first-pass, self-directed model of pitch-pair tunneling

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No outside specification was given for this model - the mechanism, the math, and the validation plan documented here are all independent design decisions, made and recorded as they were made, not reconstructed afterward. This document is the full record: what was tried, why, and what the evidence actually shows. (Originally named “Deception+” - renamed to Tunneling+ because it measures specifically pitch-pair tunneling, one of several distinct ideas that get lumped under the broader word “deception.”)

Units: distances are reported in inches (movement-scale quantities, same convention as IVB/HB elsewhere on this site); the decision point is a location along the pitch's flight path, kept in feet. **Formula:** tunneling_plus = geometric mean of THREE components - release_similarity_plus (close at release), decision_similarity_plus (close at the decision point), and plate_separation_plus (far apart at the plate) - each independently scaled 100=league average, same convention as Stuff+/wRC+. All distances are the full pairwise mean between every real pitch of type A and every real pitch of type B (not centroid-to-centroid, which understates true separation by averaging away real command variance), in inches; decision point in feet. FF/SI-involving pairs count 2x toward the pitcher-level composite (not toward the league baselines) - a hitter's read is built primarily against the fastball/sinker, so how well an offspeed pitch tunnels with THAT matters more than how two offspeed pitches tunnel with each other.

1. Executive Summary

Tunneling+ measures **pitch-pair tunneling**: how close together two of a pitcher's pitch types are at the point a hitter has to commit to swinging, versus how far apart they actually end up at the plate. A pair that looks the same early and ends up somewhere very different scores high. The position at that decision point is computed directly from Statcast's own trajectory physics - not a proxy, not a black box.

The model passed every validation test run against it:

- **Reliable**: 0.91 split-half correlation (pitchers with 2+ scoring pairs) - this is measuring something stable about a pitcher, not noise.
- **Predicts real outcomes**: positive correlation with whiff%, chase%, and swinging-strike% (r = 0.36 for swstr%), and that correlation **survives controlling for Stuff+** (partial r = 0.36) - this is not just re-deriving raw pitch quality under a new name.
- **Mechanism-specific**: in real same-at-bat sequences, pitch pairs the model scores as more tunneled produce a measurably higher whiff rate on the second pitch (z = 9.06, n = 299,712 sequences) - the specific causal pathway the model claims to capture, tested directly.

It is still a v1: the caveats and open questions in Section 7 are as important to read as the results above.

2. Why Tunneling, and Why This Isn’t Called “Deception+” Anymore

Pitching analytics converges on a few distinct mechanical ideas under the word “deception.” This model was originally named Deception+ and framed as having picked one of those ideas as its primary target - but that framing implied broader scope than the model actually has. It measures exactly one specific, well-defined mechanism, so it's named for that mechanism directly: **tunneling**.

Candidates considered when the model was first designed:

- **Tunneling** - two pitch types traveling through nearly the same point in space at the decision point, then diverging. The most commonly cited, most mechanistic definition in public pitching research, and fully computable from Statcast's trajectory fit. **This is what the model measures.**
- **Release-point consistency** - a pitcher who releases every pitch from the same spot gives no positional tell before the ball even leaves his hand. Simpler and less novel, but cheap to check independently - kept as a secondary diagnostic (Section 5), not folded into the primary score.
- **Spin mirroring** - opposing spin directions between e.g. a fastball and curveball that look similar early due to shared early Magnus-force behavior. Theoretically interesting but less rigorously validated in public work than trajectory-based tunneling - set aside for v1, noted as a real candidate for v2.
- **Effective-velocity masking** - a changeup thrown with fastball-identical arm speed. Related to tunneling but more about arm action than trajectory - not separately modeled here; to the extent it matters, it's partially captured by the release-consistency diagnostic.

Tunneling was chosen as the primary target because it is the most mechanistic (a geometric fact about two trajectories, not a proxy), the most commonly validated concept in public pitching research, and - critically - it was fully computable from data already in this database, using physics already established elsewhere in this codebase (see Section 4). The other candidates above (release-point consistency aside, which IS tracked) remain open ideas for a separate future model, not folded into this one under a broader “deception” umbrella.

3. Three Real Methodological Bugs, Caught and Fixed

3.1 Division coupled the two halves of the idea together

The first version of this model scored each pair with a single ratio: plate_distance / tunnel_distance. That coupled the two halves of the idea together by division - a pair's plate separation only counted *in proportion to* how small its tunnel separation already was. A pair with a genuinely large plate separation (literally the point of throwing two different pitch types) could still score below average purely because its tunnel separation was also moderately large - real, meaningful separation was being diluted rather than credited. Caught via a real example: Tarik Skubal's changeup/fastball pair had the single largest plate separation in his entire arsenal, yet scored a below-average score under the ratio, purely because its tunnel separation was also middling - the formula was punishing exactly the thing two different pitch types are supposed to do.

Fix: split into independently-normalized components, each scaled 100=league average (decision_similarity_plus and plate_separation_plus at first; see 3.3 for the third), combined by **geometric mean** instead of division. Geometric mean still requires every component to be reasonably good to produce a high combined score - a pair that's merely average on one axis can't be “rescued” to a great score by another axis alone. This preserves the AND-logic that's the actual definition of tunneling, without the original ratio's harsh division-based coupling.

3.2 Centroid distance quietly erases real command variance

Every distance was originally computed as centroid-to-centroid: average all of a pitcher's changeups into one point, average all his fastballs into one point, then measure the gap between those two averages. That silently assumes away pitch-to-pitch variance within each type - checked directly on Skubal's changeup/fastball pair: the two types' centroids sit 5.3" apart at the decision point, but each type's own individual pitches vary by 4-5" (std. dev.) around their own centroid. The *typical real* changeup and *typical real* fastball are actually 9.7" apart at that moment - not 5.3". Centroid distance was systematically understating true separation for any pitcher with real command variance, which is to say every real pitcher. Leaguewide, average tunnel distance nearly tripled once measured correctly (3.4" $\rightarrow$ 9.5").

Fix: every distance (release/tunnel/plate) is now the full pairwise mean distance between EVERY individual pitch of type A and EVERY individual pitch of type B, not the distance between their averages. Computed via vectorized broadcasting rather than a Python loop over pitch pairs - cheap even at full-season pitch counts (roughly 15 seconds extra, leaguewide, for a materially more honest number).

3.3 A release-point component was tried, tested, and rejected

A pair could converge beautifully by the decision point yet still tip itself off at release - a changeup thrown from a visibly different arm slot than the fastball gives the hitter information before the ball is even 10 feet from the pitcher's hand, regardless of how similar the two pitches look later. release_distance was always computed and displayed, but never fed into the score - so a release_similarity_plus component was built (100 × league-average release distance / this pair's release distance) and tried as a third factor in the geometric mean alongside decision_similarity_plus and plate_separation_plus.

The theory was reasonable; the data disagreed. Tested against the full validation battery, adding release_similarity_plus made the model measurably WORSE, not better: the sequential same-at-bat test - the single most rigorous, mechanism-specific check available (Section 6.3) - dropped from z=9.06 (two components) to z=1.60 (three components), falling below the conventional significance bar entirely. Every outcome correlation weakened similarly. Real hitters' whiff rates on the second pitch of a sequence just don't track release-point similarity the way they track decision-point similarity and plate separation - folding it in diluted a strong, validated signal with a much weaker one.

Decision: rejected. release_similarity_plus is still computed and shown per-pair in the arsenal inspector, for context - but tunneling_plus stays the two-component geometric mean from 3.1. This is a real instance of the working principle for every model on this site: a plausible mechanism gets tested against the evidence before it ships, and loses to the evidence when the two disagree.

Every fix in this section is a genuinely different scoring function from what came before, not a rescaling - pair rankings change, not just their units. Every validation check in Section 6 was re-run against the current (two-component, full-pairwise-distance) version rather than assumed to still hold from an earlier one.

4. The Physics

Statcast fits every pitch to a constant-acceleration trajectory: velocity (vx0, vy0, vz0) and acceleration (ax, ay, az) are given at a fixed measurement plane 50 feet from home plate. This codebase already uses that exact model elsewhere - metrics.py's compute_vaa (Vertical Approach Angle) solves for the ball's state at the front of the plate given its state at y=50ft. This model generalizes that same solve two ways: to BOTH the x and z axes (not just z), and to an arbitrary target distance from the plate (not just the plate itself). The same simplification compute_vaa already makes - treating release_pos_x/release_pos_z as the position at the y=50ft plane, when release actually happens a few feet closer - is kept here for consistency with the rest of the codebase rather than introducing a second convention.

Decision point: 23.5 feet from home plate. Public pitch-tunneling research generally cites a range of roughly 20-24 feet from the plate for when a hitter's swing decision is effectively locked in, depending on pitch velocity. This model uses one fixed value inside that range for every pitch - deliberately simple for a v1. A velocity-adjusted decision point (faster pitches leave less flight time, so the true decision point should sit slightly further from the plate) is a real, documented limitation, not an oversight - see Section 7.

For every pitch, three positions are computed or read directly:

- **Release** - release_pos_x / release_pos_z, read directly from Statcast.
- **Decision point** - position at y = 23.5 ft, solved from the trajectory equations (the new part).
- **Plate** - plate_x / plate_z, read directly from Statcast.

5. The Pipeline, Step by Step

1. Load every pitch for the season with a valid trajectory (release position, velocity, acceleration, and plate location all present) and compute its decision-point position.
2. Group by (pitcher, pitch type) to identify which types qualify - at least 100 pitches that season (a type thrown only a handful of times is too thin a sample to trust).
3. For every pair of qualifying pitch types within one pitcher's arsenal, compute three distances - release distance, tunnel distance (at the decision point), and plate distance - each as the FULL PAIRWISE MEAN distance between every individual pitch of type A and every individual pitch of type B (see Section 3.2), not a distance between averaged positions.
4. Compute the league-wide, usage-weighted average release/tunnel/plate distance across every qualifying pair, leaguewide - the baselines every pair is scored against (each is floored at 1.8" first, so a near-zero separation can't produce an explosive, meaningless component).
5. For each pair, compute release_similarity_plus, decision_similarity_plus, and plate_separation_plus (each 100 × the relevant ratio against its league baseline). tunneling_plus = the geometric mean of decision_similarity_plus and plate_separation_plus only - release_similarity_plus is shown per-pair for context but is not scored; it was tested as a third component and rejected after it measurably hurt validated predictive power (see Section 3.3).
6. Weight each pair by the geometric mean of the two types' usage share within the pitcher's qualifying arsenal - a pair only matters to a hitter's real experience in proportion to how often both pitches actually show up. Pairs involving the four-seam (FF) or sinker (SI) then get a further 2x bonus on top of that usage weight before aggregating to the pitcher-level composite: a hitter's read is built primarily against the fastball (or sinker, for an east-west arsenal), so how well an offspeed pitch tunnels with THAT matters more than how two offspeed pitches tunnel with each other. This bonus does not apply to a pair's own tunneling_plus (shown per-pair in the inspector) or to the league-wide baselines - only to how much each pair counts toward the single pitcher-level number.
7. Aggregate to one tunneling_plus per pitcher: the (usage x anchor-bonus)-weighted average of tunneling_plus across every real pair in their arsenal - the headline number shown on the site.
8. Separately compute release_consistency: the usage-weighted RMS distance of each individual pitch's release point from the pitcher's own overall release centroid, across the WHOLE arsenal (not just pairs) - a simpler, whole-arsenal diagnostic, distinct from the pairwise release_similarity_plus above, kept separate from the primary score.

6. Validation

Four independent checks were run against the model - not selected after the fact, run as a deliberate battery because a single passing test proves very little on its own.

6.1 Split-half reliability

Each pitcher's pitches were split into two random, independent halves; the score was computed separately on each half and correlated. A metric dominated by noise would show a correlation near zero.

Cutoff	n pitchers	Correlation
min 1 pair(s)	369	0.897
min 2 pair(s)	206	0.914
min 3 pair(s)	206	0.914

6.2 Does it predict real outcomes?

tunneling_plus was correlated against each pitcher's real 2025 outcomes (n = 546 pitchers with 300+ pitches). To check this isn't simply re-deriving pitch quality under a new name, the correlation with this project's own Stuff+ v2 model was checked first (r = 0.110 - modest, not the 0.6+ that would suggest heavy overlap), then each outcome's correlation with tunneling_plus was recomputed AFTER removing whatever Stuff+ already explains (linear residualization).

2025 outcome	Raw r	Partial r (Stuff+ removed)
whiff_pct	+0.304	+0.289
chase_pct	+0.345	+0.329
swstr_pct	+0.362	+0.361
k_minus_bb_pct	+0.223	+0.195
xwoba	-0.128	-0.081

Every outcome except xwOBA shows a meaningful positive partial correlation - tunneling adds real signal beyond what raw pitch quality already predicts, for the swing-and-miss family of outcomes especially (swinging-strike% partial r = +0.361). xwOBA's partial correlation flips sign and stays weak - plausible, since tunneling should matter most for swing decisions (whiff/chase), less for the quality of contact once a batter does connect.

6.3 Sequential same-at-bat test

The most mechanism-specific check: within real at-bats, when a pitcher throws two DIFFERENT pitch types back to back, does the pre-computed tunneling_plus for that specific pair predict a worse outcome on the second pitch of the sequence? This tests the actual causal pathway the model claims to capture, not a league-wide aggregate that could be confounded by “good pitchers are good at everything.”

Tunneling+ tercile	Whiff rate (2nd pitch)	n sequences
Low	11.6%	99,926
Mid	11.7%	100,024
High	12.9%	99,762

Low-to-high tercile difference: z = 9.06 (|z| > 1.96 is the conventional p < 0.05 bar). The effect is small in absolute terms (a percentage point or so of whiff rate) but, given the enormous sample (n = 299,712 sequences), highly statistically significant and monotonic across all three terciles.

6.4 Worked examples

A soft, secondary sanity check (not a substitute for the tests above) - real pitcher names at the extremes, to see whether the ranking is at least plausible against general scouting narratives. Treat this section as illustrative, not as independent statistical evidence - I have not independently verified every pitcher's specific reputation against outside sources.

Highest tunneling_plus:

Pitcher	Tunneling+	Pairs	Pitches
Tony Santillan	108	3	1290
Justin Sterner	107	3	1104
Spencer Strider	107	3	2013
Emmet Sheehan	107	3	1221
Cade Smith	107	3	1257

Lowest tunneling_plus:

Pitcher	Tunneling+	Pairs	Pitches
Yohan Ramírez	89	3	566
Matt Strahm	90	6	1018
Anthony Bender	90	3	764
Joey Cantillo	91	6	1717
Carmen Mlodzinski	92	15	1710

7. Known Limitations and Open Questions

- **Fixed decision point.** 23.5 ft from the plate is used for every pitch regardless of velocity. A 100mph fastball and a 75mph curveball don't take the same time to travel the same distance - a velocity-adjusted decision point (further from the plate for faster pitches, since less flight time remains) would be more physically honest. Not implemented in v1; a clear v2 candidate.
- **n_pairs=1 pitchers are noisier.** A pitcher with exactly two qualifying pitch types has only one pair to average - no internal averaging to smooth out noise the way a 6-pair pitcher's score benefits from. The worked examples in Section 6.4 were filtered to n_pairs≥2 for this reason; the full leaderboard (in the HTML page) is not, matching the same trade-off documented for Stuff+ v2's small-sample pitch types.
- **No shrinkage applied.** Unlike Stuff+ v2, this model does not (yet) apply empirical-Bayes shrinkage toward league-average component values for thin samples. Given the demonstrated split-half reliability is already strong, this is a lower priority than it was for Stuff+, but remains untested at very low pair counts.
- **Spin mirroring not modeled.** A pitch pair with genuinely mirrored, opposite spin axes may look more similar to a real hitter's eye than pure trajectory distance alone would suggest, or vice versa - this model has no visibility into that.
- **Correlational, not causal, at the aggregate level.** Section 6.2's outcome correlations are exactly that - correlations. Section 6.3's sequential test is the strongest causal-adjacent evidence in this report specifically because it isolates the mechanism directly, but even that is observational, not a controlled experiment.
- **Single season.** All validation here is within the 2025 season. No year-over-year predictive-validity check (does a pitcher's tunneling_plus in one season predict a DIFFERENT season's outcomes) has been run yet - the same gold-standard test that mattered for validating Stuff+ v2.

8. Suggested Next Steps

- Velocity-adjust the decision point per pitch instead of a fixed 23.5 ft.
- Year-over-year predictive validity check, matching the Stuff+ v2 methodology.
- Explore spin-axis mirroring as a second, independent signal alongside trajectory tunneling.
- Apply empirical-Bayes shrinkage to stabilize thin-sample (especially n_pairs=1) pitchers, matching the approach already validated for Stuff+ v2.
- Combine tunneling_plus and release_consistency into one blended metric, once/if each is shown to add independent value in a joint model rather than kept separate.
- Revisit release-point information in a different form than a third geometric-mean factor (Section 3.3) - e.g. as a separate filter/flag on obviously-tipped pairs, or weighted much lighter than the other two components - rather than abandoning the idea outright just because the naive equal-weight version didn't validate.